

ADB INSTALL AUR SETUP (ANDROID DEBUG BRIDGE) – STUDY NOTES

By: SYED GHOU

ADB KYA HAI?

ADB (Android Debug Bridge) ek command-line tool hai jo computer aur Android device ke darmiyan connection banata hai. Is ki madad se computer se Android device par commands run ki ja sakti hain, apps install ki ja sakti hain, files transfer ki ja sakti hain aur debugging ki ja sakti hai.

ADB SETUP KYA HAI?

ADB Setup ka matlab hai computer par ADB environment ko install aur configure karna taake Android device ke sath communication establish ho sake. Setup complete hone ke baad user **ADB commands** use kar sakta hai.

ADB SETUP KE LIYE ZAROORI CHEEZEN

- Computer (Windows/Linux/macOS)
- Android SDK Platform Tools
- Android device



- USB cable
- USB Drivers (Windows mein zaroorat ho sakti hai)
- USB Debugging enabled phone

ADB SETUP KARNE KA TARIKA__STEP-BY-STEP DETAIL

1. PLATFORM TOOLS INSTALL KARNA

Sab se pehle Android SDK Platform Tools download karke computer mein extract karein. Isi folder mein ADB ki files hoti hain.

2. DEVELOPER OPTIONS ENABLE KARNA

Phone mein:

Settings → About Phone → Build Number

par 7 dafa tap karein. Is se **Developer Options** activate ho jate hain.

3. USB DEBUGGING ON KARNA

Developer Options mein ja kar:

USB Debugging → ON

karein. Yeh option computer ko Android device ke sath communicate karne ki permission deta hai.

4. DEVICE CONNECT KARNA

Phone ko USB cable se computer ke sath connect karein. Phone par agar popup aaye:

"Allow USB Debugging?"

to **Allow** select karein.

5. CONNECTION CHECK KARNA



Command Prompt / Terminal open karein aur command likhein:
</> BASH

adb devices

Agar setup sahi ho to output kuch is tarah hoga:

List of devices attached

ABC123456 device

ADB INSTALL KYA HAI?

ADB Install ek command hai jis se computer se Android device par APK file install ki jati hai. Developers aur cyber security professionals isay app testing aur security analysis ke liye use karte hain.

ADB INSTALL COMMANDS

NORMAL APK INSTALL

</> BASH

adb install app.apk

APP UPDATE KARNA

</> BASH

adb install -r app.apk

Purpose: Existing app ko uninstall kiye baghair update karta hai.

PERMISSIONS KE SATH INSTALL

</> BASH



adb install -g app.apk

Purpose: Testing ke liye app ki runtime permissions automatically allow karta hai.

APP UNINSTALL KARNA

</> BASH

adb uninstall package_name

Purpose: Installed application remove karta hai.

CYBER SECURITY MEIN ADB KI IMPORTANCE

Cyber Security mein ADB ka use Android applications ko test karne, vulnerabilities find karne aur malware analysis ke liye hota hai. Ethical hackers ADB ke through apps install karke unki permissions, files, logs aur behavior analyze karte hain.

ADB KE FAYDE

- APK files manually install kar sakte hain.



- App testing aur debugging easy hoti hai.
- Device logs check kiye ja sakte hain.
- Files transfer ki ja sakti hain.
- Android security testing mein help milti hai.

Security Risks

Agar USB Debugging hamesha ON ho ya unknown APK install ki jaye to security risk barh sakta hai. Malicious APK device mein malware install kar sakti hai ya personal data steal kar sakti hai.

PRO TIP

💡 **ADB use karne ke baad USB Debugging OFF kar dein aur sirf trusted APK files install karein taake Android device secure rahe.**

NOTE;

THESE NOTES ARE ONLY FOR STUDY PURPOSES

AUTHOR:

Syed Ghous

